DESTROYED CITY GENERATOR

PROCEDURAL CITY GENERATION TOOL



User Guide

1. Overview

The Procedural Destroyed City Generator is an editor-only tool that automatically arranges the buildings, rubble and debris from this pack into a complete ruined city. You assign the catalog, press Generate, and it lays out streets, districts, building blocks and scattered debris for you.

> **Important:** The generator is meant to give you a strong starting point. For the best-looking and unique results, treat the generated city as a base and then customize it by hand, nudge hero buildings, clear or add areas, fix any awkward overlaps, and dress the scene with your own lighting, fog and set pieces.

2. Quick Start

- Open the generator: Tools ▸ **Destroyed Buildings and Debris Pack ▸ Destroyed City Generator.**
- Assign the preconfigured catalog in the window's Catalog field, you can find it in the same folder as the demo.
- If you are using a terrain and want the streets to be painted automatically, you will need to configure a few additional options in the generator, please refer to **Section 5.11** before continuing.

- Select the ground mask (section 5.12).
- Press Generate.

The city will appear in the world origin (location 0,0,0) under a single parent object (named "Placed City" by default). Press Reroll + Generate for a different random variation, or Clear to remove it.

2.1 Suggested Workflow

- **Create a new scene.**
- **Add and center a terrain.** Create two terrain texture layers: the default layer and another one for the streets.
- **Set up the lighting.** Copy the lighting and post-processing volume from the existing demo scene.
- **Configure the generator.** Open the generator, assign the catalog, and configure the street texture layer as described in **Section 5.11**.
- **Generate the city.** The generated city may not appear directly in front of the camera. Look around the Scene view or locate it in the Hierarchy and double-click it to focus the Scene view on the generated city.
- **Start small.** Experiment with the available parameters and gradually increase the city size as needed. This makes it easier to understand how the different settings affect the result and helps avoid generating unnecessarily large scenes while testing.
- **Customize and refine the generated city.** Use the generated layout as a starting point and manually adjust, replace, or add elements to achieve the best end results.

3. The Catalog

The catalog is a small asset that stores which prefabs to use, what each one is, and all of its settings. It's already preconfigured with the prefabs from the pack, but you can extend it by adding your own assets if needed.

When extending the catalog, keep performance in mind. Any assets you add should be sufficiently optimized to be placed potentially thousands of times throughout the scene.

Every item in the catalog appears in a list, grouped by Kind, each with a preview thumbnail and these controls:

- **Enabled (checkbox) -** untick to keep a prefab in the catalog but exclude it from generation.
- **Kind -** Building / Rubble / Small Debris / Street Debris / Ignored.
- **Size -** Small / Medium / Large / Skyscraper.
- **Tilt -** whether the generator may lean this piece a few degrees.
- **Weight -** how likely this piece is to be chosen relative to others of the same kind. 2 means twice as likely as a piece weighted 1.

4. The Generator Window

Open it from **Tools ▸ Destroyed Buildings and Debris Pack ▸ Destroyed City Generator**.

At the top: the Catalog field, a status line showing how many pieces are placeable and how many are buildings, an Edit Catalog button, and Expand all / Collapse all for the settings sections.

The settings are organized into collapsible sections (described in full in the next section). At the bottom is the action bar:

- **Generate** - build the city. Removes any previous city of the same name first.
- **Reroll + Generate** - pick a new random seed and generate a fresh variation.
- **Clear** - remove the generated city.

5. Parameters

All parameters live on the catalog and are edited from the generator window, grouped into the sections below.

5.1 General

- **Seed** - the random number that drives the whole layout. The same seed with the same catalog and settings always produces the exact same city. Change it (or use Reroll + Generate) to get a different variation.
- **Root Object Name** - the name of the parent GameObject the whole city is created under. Clear removes the object with this name, so if you want to keep several generated cities in one scene, give each a different name before generating.
- **City Size (X, Y)** - the width and depth of the whole city area, in world units. The city is generated starting at the world origin and extending to this size. Bigger values = a bigger city and (a lot) more objects.

5.2 City shape

By default the city fills the whole City Size rectangle, so its outline is a hard square or rectangle. Turn this on to give it a natural, irregular outline instead.

- **Organic Shape** - the toggle. Off = the full square/rectangle (unchanged from before). On = the irregular outline described here. Everything else in this section only matters when it's on.
- **Organic Edge** - how far the outline bulges in and out. `0` = a clean rounded ellipse; higher = lumpier, more ragged edges with deeper inlets and bays.
- **Organic Lobes** - how many bumps run around the outline. Low values give a few big lobes; high values give many small inlets.

The outline is driven by the Seed, so Reroll + Generate gives a new shape as well as a new layout. Streets, debris and terrain-street painting all follow the outline, the ground outside it is left bare rather than paved or littered. The optional Background Ring still forms a rectangle around the city area, so it reads as a distant skyline enclosing the organic city.

5.3 Streets & Blocks

The city is divided into rectangular blocks separated by streets; buildings are packed inside the blocks.

- **Block Size (min, max)** - the smallest and largest a city block can be. Each block gets a random size in this range. Set the maximum above your tallest building's footprint, otherwise the widest towers won't fit inside a block and downtown will fill with smaller buildings instead.
- **Side Street Width** - width of ordinary streets between blocks.
- **Main Street Width** - width of the wider main avenues.
- **Main Street Every** - every Nth street is a wide main avenue instead of a side street. 4 means every fourth street is a main avenue.
- **Block Setback** - an empty margin kept just inside each block's edge, so buildings don't spill out into the road.
- **Building Gap** - the gap left between neighbouring buildings packed inside a block.

5.4 Organic Streets

These break up the regular grid so the street network doesn't look perfectly rectangular.

- **Block Merge Chance** - the chance that a block "swallows" the street next to it and merges with its neighbour. This produces irregular block sizes, T-junctions and dead-end streets. 0 = a clean grid; higher = more irregular.
- **Avenue Count** - how many long, winding avenues are cut diagonally across the whole city on top of the grid. These add curves and diagonals a pure grid can't produce, and carve building-free corridors along their path (this happens whether or not you paint the terrain).
- **Avenue Width** - how wide those avenues are.
- **Avenue Curve** - how sharply the avenues bend as they travel, in degrees per step. 0 = straight diagonals; higher = winding, snaking roads.
- **Avenue End Chance** - the chance, per step, that an avenue simply stops in the middle of the city, leaving a dead end. Keep it small.

5.5 Districts

Blocks are assigned to districts by how close they are to the downtown centre, so the skyline rises toward the middle like a real city. There is always at least one downtown block, no matter how small the city is.

- **Downtown Center (X, Y)** - where the tall core sits, as a fraction of the city. (0.5, 0.5) is the middle.
- **Downtown Radius** - roughly what share of blocks become downtown (filled with big buildings). Raise it for a bigger tall core.
- **Midtown Radius** - the share of blocks that become midtown (medium and large buildings). The rest become outskirts (small buildings and slums).
- **Zone Noise Scale** - how fine the noise is that makes the district borders ragged instead of perfect rings.
- **Zone Noise Amount** - how strongly that noise pushes the district borders around. 0 = clean concentric districts; higher = more irregular, natural borders.
- **Downtown Density** - how many of a downtown block's plots actually get a tower (1 = a tower on every plot).
- **Outskirts Tall Chance** - the chance that a low-rise block instead holds a single tall building — a lone tower surviving among the small buildings. 0 = never.

5.6 Buildings

- **Density** - how full each ordinary (non-downtown) block gets packed. Lower it to leave courtyards and open gaps; raise it for a denser, tightly packed city. (Every block still gets at least one building.)
- **Align To Grid** - when on, buildings are rotated in clean 90° steps (a tidy, city-block look). Turn it off to let buildings face any angle. (Debris is always randomly rotated regardless of this setting.)
- **Tilt Max Degrees** - the maximum lean applied to buildings whose Tilt is enabled (the big ones). A little tilt makes tall ruins look damaged and unstable.
- **Building Scale Jitter** - a small random size variation applied to each building, up to this fraction. Buildings are only ever scaled up (never shrunk), so they never look smaller than intended. Capped at 10%.
- **Tilt Sink Per Degree** - how far a tilted piece is sunk into the ground for each degree it leans, so its lifted side doesn't float. 0.25 means it sinks 1 unit for every 4° of lean. (See Grounding below.)

- **Ground Embed** - a tiny amount every piece is sunk below the ground so nothing hovers due to floating-point precision. Usually left small.

5.7 Clustering

Clustering groups the same type of building together into neighbourhoods, so the city reads as having distinct areas rather than a random jumble.

A family is a main model plus its variants: building6, building6-variant1 and building6-variation2 all belong to family building6, while building7 is a separate family.

- **Cluster Buildings** - turn same-type grouping on or off. Off = every block mixes all building types.
- **Cluster Zone Size** - how large a same-family neighbourhood is, in world units. Bigger = larger areas of one type.
- **Cluster Strength** - how strictly a block sticks to its family. 1 = only that family; lower values mix in other types (0.65 = mostly one dominant type with some variety).

5.8 Debris

- **Rubble Per Big Building** - how many large rubble piles are clustered around each big building. Big rubble is only placed near big buildings, never scattered across the map.
- **Rubble Spread** - how far out from a big building's footprint the rubble can land.
- Small Debris Per Building - small debris is placed around every building. Because it's a fixed count per building, small buildings end up with denser litter than big ones (whose count spreads over a larger footprint).
- **Small Debris Spread** - how far small debris spreads from a building, as a multiple of the building's footprint radius.
- **Small Debris Per Area** - small debris scattered evenly across the open ground (courtyards, empty blocks) per 100 square units, so nothing looks barren. This never lands on streets.
- **Street Debris Per Area** - street debris (pieces you set to the Street Debris kind) scattered along the streets and avenues, per 100 square units.

Small debris and street debris are kept on separate surfaces on purpose: small debris fills the ground around buildings, street debris litters the roads.

5.9 Debris Scale

Each debris type has its own random size range, for variety.

- **Debris Scale Min / Max** - the smallest and largest random scale for small debris (1 = the model's authored size).
- **Rubble Scale Min / Max** - the smallest and largest random scale for rubble.

5.10 Background Ring

Optionally surrounds the city with distant background buildings so it feels like it continues beyond the generated area.

- **Background Enabled** - turn the background ring on or off.
- **Background Depth** - how far out from the city edge the ring of background buildings extends.
- **Background Density** -how full the ring is.
- **Background Gap** - a clear margin between the city edge and where the background ring starts.
- **Background Rubble Share** - how much of the ring is collapsed rubble versus standing skyscrapers.

The background pieces are placed under a separate "Background" child object so you can hide or delete them independently.

5.11 Terrain Streets

If your scene has a Unity Terrain, the generator can paint the streets onto it automatically.

- **Paint Terrain Streets** - turn street painting on. A Terrain field appears in this section when it's on — leave it empty to use the scene's active terrain.
- **Street Layer** - which of the terrain's existing Terrain Layers represents the road/sand surface. This layer must already be added to the terrain (in the terrain's Paint Terrain settings).

- **Terrain Streets Swap XZ** - only needed if the painted streets come out rotated 90° from the buildings; ticking it swaps the terrain's X/Z splatmap axes to line them up.
- **Street Edge Wobble** - how far the painted street edges wander off the grid, for an organic, weathered look. 0 = perfectly straight edges.
- **Street Noise Scale** - how busy that edge variation is (higher = smaller, tighter wobbles).
- **Street Strength Min** - the lowest strength the street layer is painted at, so the road fades toward the underlying sand in patches. 1 = solid streets everywhere; lower = more worn and broken.

Notes on painting:

The city is generated at the world origin over the City Size area, so position and size your terrain to cover that region — painting is skipped outside it.
Higher splatmap resolution on the terrain gives crisper streets.
The street layer must already exist on the terrain, or the generator logs a warning and skips painting.

5.12 Ground

- **Ground Y** - the flat height everything is placed at when Conform To Ground is off.
- **Conform To Ground** - instead of a flat plane, raycast straight down onto existing colliders (terrain or meshes) and place each piece at the ground height it finds.
- **Ground Mask** - which layers the ground raycast is allowed to hit.

5.13 Output

- **Mark Static** - mark every generated object as static (for batching, lightmaps, occlusion). Leave this off while iterating — turning it on makes Unity build static-batch meshes the first time you click in the Scene view, which freezes the editor for a while on a big city. Turn it on only once, on a scene you're finalising and keeping.
- **Objects Per Frame** - how many objects are placed each editor frame. The city is built in batches so the editor stays responsive and shows a progress bar. Lower this for a smoother editor while building; raise it to finish faster.